AI-Use Disclosure Statement

IBM Qiskit Fall Fest 2026 Challenge · QC@IITI, Indian Institute of Technology Indore

Project: Quantum Traffic Gridlock Prototype
Author / Lead: Harsh Parmar
Track: Urban Signal Gridlock Optimization
Execution: 100% Offline CPU (Qiskit 2.x)
Repository: github.com/harshparmar2004/Quantum-Traffic-Gridlock-Prototype
Certification: Academic Integrity & Open Science

1. DECLARATION OF AI ASSISTANCE & UTILIZED MODELS

In accordance with the academic integrity and open science guidelines of the IBM Qiskit Fall Fest 2026, the author formally declares that Generative AI assistants—specifically **Google DeepMind Antigravity** and **Anthropic Claude 3.5 Sonnet**—were utilized as software engineering accelerators and technical documentation aids during the development and benchmarking of this project.

2. DETAILED TASK BREAKDOWN: AI ASSISTANCE VS. HUMAN AUTHORSHIP

To ensure complete transparency, the following matrix delineates the exact scope of AI utilization versus independent human scientific research, mathematical derivation, and implementation:

Development Domain	AI Tool Assistance	Human Author Contributions	Authorship
QUBO & Ising Modeling	LaTeX equation syntax checks & table formatting	Conceived continuous fluid queue conservation law, Lyapunov drift objective (Q^2), and V2 demand-capping	**100% Human**
Quantum Algorithm (QAOA)	OpenQASM 3 gate serialization syntax helpers	Designed p=3 Fourier parameterization (q=2), native *RZ/RZZ/RX* circuits, and multi-restart COBYLA	**100% Human**
Section 11 Innovations	Proofreading technical prose and notation	Derived 12-qubit corridor cascade, MUTCD Laplacian safety penalty, and Richardson ZNE routines	**100% Human**
Test Suites & Verification	Pytest parameterization boilerplate code	Wrote 22 test fixtures, verified 4 official test vectors, and conducted 300-scenario regret calibration	**100% Human**
Software CLI & Server	Drafted standard library argument parser & HTTP routing	Integrated simulation dynamics, solver API, error handlers, and programmatic `solve(...)` interface	**Assisted**
Operations Visualizer	CSS Grid structure & canvas circle coordinate helpers	Designed cloud-beige UI/UX, phase animation logic, queue bubbles, and real-time telemetry streaming	**Assisted**

3. PROMPTING METHODOLOGY & VERIFICATION PROTOCOL

AI tools were queried interactively with strict human-in-the-loop oversight. Prompts were confined to boilerplate scaffolding, schema verification, and layout generation (e.g., "*Format this matrix as a clean HTML table*" or "*Write a standard-library HTTP handler with CORS headers*"). **Zero automated code was merged without rigorous unit test verification and mathematical proof.** All algorithmic outputs were validated against ground-truth exact state enumeration (N = 256) and official verification vectors.

4. STATEMENT OF ORIGINALITY & COMPETITION COMPLIANCE

The author certifies that no proprietary solvers, unauthorized external packages (e.g., `qiskit-aer`), or pre-computed schedule lookup tables were employed. The submission executes entirely offline on CPU architectures, achieves 0.00% surrogate regret on Reference Scenario R and a quantum Lift of 0.6389, and fully adheres to all competition rules and ethical standards.

Certified by: Harsh Parmar | **Affiliation:** QC@IITI, Indian Institute of Technology Indore
Git Commit: `b2ea377` | **Date:** October 2026

VERIFIED & CERTIFIED SUBMISSION